

Class 12:

1. Using double integrals find moment of inertia about the x-axis of the area enclosed by the lines

$$x = 0, y = 0, \frac{x}{a} + \frac{y}{b} = 1$$

Ans: $\frac{ab^3}{12}$

2. Find the moment of inertia of an octant of ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ about x-axis.

Ans: $\frac{abc(b^2 + c^2)\pi}{30}$
